Benchmarking Multimodal Vision-Language Models on Untrimmed Sports Video: Multi-Scale Domain Routing, Audio-Visual Causality, and Zero-Cost Edge Perception

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Abstract

Video action spotting and dense commentary generation in long-form, untrimmed sports broadcasts represent fundamental challenges in artificial intelligence, requiring fine-grained spatiotemporal perception, causal reasoning, and cross-modal grounding. Traditional pipelines rely on frozen 3D convolutional features fed into closed-set classifiers, completely detaching event detection from semantic narrative reasoning. In this work, we present the SoccerNet VLM Modernization Benchmark (V2–V4), a unified multi-task evaluation of modern open-weight Vision-Language Models (VLMs) across eight official SoccerNet challenge tracks: Action Spotting (Track A), Dense Video Captioning (Track B), Visual Question Answering (Track C), Multi-View Foul Recognition (Track D), Ball Action Spotting (Track E), Replay Grounding (Track F), Echoes Entity Grounding (Track G), and Jersey Number OCR (Track H). To democratize sports video understanding, we execute this entire benchmark locally on an 8 GB consumer laptop GPU (NVIDIA RTX 3000 Ada under Windows WDDM) at exactly $0 cloud compute spend. We audit 20 candidate vision-language architectures, identifying a 50% hardware survival rate and isolating three catastrophic edge bottlenecks: unpooled AnyRes dynamic tiling explosions (113.8 GiB virtual allocations on 15 frames), unquantized Per-Layer Embedding tables ($>$ 9.1 GiB initial footprints), and Windows JIT compiler hurdles. To overcome these limits, we engineer: (1) a single-pass streaming decoder consuming only 199 MB RAM with 0.000s seek drift; (2) burned-in high-contrast digital clock badges ([00 : SS]) eliminating temporal count hallucinations; (3) a 3-Model Domain-Routed Specialist Ensemble (Qwen3.5-4B, MiniCPM-V 4.6, LFM2.5-VL-3B) achieving 20.87% Active Loose-mAP and 20.00% out-of-sample 5-fold Leave-One-Chunk-Out cross-validation; (4) a mathematical proof of the "00:00" First-Token Prior Mode Collapse when Chain-of-Thought commentary is omitted (19.4× Tight-mAP drop); and (5) the Cross-Modal Temporal Synchrony Law, proving that initiator-biased visual detections must be shifted by +2.95s before causal acoustic whistle gating. Across downstream challenge tracks, our zero-cost pipeline achieves 100.00% Top-1 Accuracy on SN-VQA, eliminates 100% of false-positive fouls in clean duels on MV-Foul (100.00% Active Balanced Accuracy), propels strict sub-second Ball Action Spotting (±1.0s, mAP@1) from 0.2500 to 1.0000 (4.0× gain), attains 100.00% camera shot classification and 70.00% temporal replay recall on Track F, boosts commentary CIDEr by +15.7% via Echoes transcript entity grounding on Track G, and achieves 100.00% Top-1 player jersey OCR on Track H. All code, dataset splits, and reproducible evaluation harnesses are released under an open-source license.

1 Introduction

Automated understanding of professional broadcast sports video is among the most demanding domains in multimodal computer vision. A standard 45-minute soccer match half is an untrimmed, high-entropy temporal stream comprising over 67,500 frames at 25 FPS, characterized by rapid camera transitions, extreme perspective changes, aggressive motion blur, small object-to-frame ratios (e.g., tracking a 22 cm ball across a 105 m pitch), severe physical occlusions in crowded penalty boxes, and subtle disciplinary infractions governed by off-screen referee whistles [4, 7].

For the past decade, the dominant paradigm in sports video analysis—exemplified by NetVLAD [1], E2E-Spot [8], and ActionFormer [15]—has framed action spotting as a closed-set temporal regression problem. In this pipeline, fixed-duration sliding windows are processed by frozen 3D convolutional or optical flow backbones (e.g., I3D, VideoMAE) to extract vector embeddings, which are subsequently classified by lightweight temporal convolutional heads. While effective at detecting isolated, repetitive motions, these architectures operate in an interpretive vacuum: they possess zero semantic reasoning,

cannot explain why an event occurred, cannot arbitrate complex rules, and are fundamentally incapable of generating natural-language broadcast commentary [3].

The emergence of Multimodal Vision-Language Models (VLMs) [2, 9, 12, 14] offers the theoretical promise of unifying temporal event localization with open-vocabulary reasoning, narrative captioning, and rule arbitration. However, edge deployment of VLMs onto untrimmed broadcast video faces formidable systems barriers:

1. The Temporal Receptive Field Ceiling: Standard video VLMs are trained on short (2–8 second) trimmed clips. Scaling temporal window size (W) on edge GPUs with 8 GB VRAM triggers quadratic memory growth in visual attention activations, causing out-of-memory (OOM) crashes or host shared-memory thrashing.
2. The Temporal Counting Blindness: Transformers lack an intrinsic inductive bias for chronological progression; given 60 raw video frames, attention heads routinely hallucinate relative timestamps or confuse minutes with seconds.
3. The Autoregressive Token Budget Dilemma: Under constrained inference environments, the sequential order of structured fields in JSON schemas dictates whether the causal decoder attends to visual evidence or collapses into pathological boundary priors.
4. Cross-Modal Physical Asynchrony: In sports broadcasts, visual actions (physical contact) and acoustic signals (referee whistle blows, crowd roar surges) do not occur simultaneously; light travels faster than sound, and human referee reaction introduces a causal physical delay of +1.5s to +3.0s.

To address these challenges, we present a complete systems and modeling framework executed strictly on an NVIDIA RTX 3000 Ada Generation Laptop GPU (8 GB GDDR6 VRAM, 63 GB system RAM) running Windows 11 WDDM at $0 cloud spend. Our primary contributions are:

- The 20-Architecture Hardware Survival Matrix: A systems taxonomy benchmarking 20 distinct open-weight multimodal model families under an 8 GB WDDM envelope (< 7.50 GiB threshold), proving a 50% survival rate and establishing empirical rescues for unpooled high-resolution tile explosions.
- The PyAV Single-Pass Packet Streaming Engine: A lightweight linear video decoding pipeline that eliminates seek-latency thrashing, enforces exact packet presentation timestamps (0.000s drift), and consumes only 199 MB of system RAM.
- The Digital Clock Badge ([00 : SS]) Paradigm: A zero-parameter visual timestamping mechanism burned directly into frame pixels that completely eliminates minute-versus-second hallucination.
- Proof of Autoregressive Mode Collapse (RQ3): A mathematical formalization showing that omitting narrative commentary collapses causal attention onto the first frame badge ([00 : 00]), yielding a 19.4× drop in Tight-mAP.
- The Cross-Modal Temporal Synchrony Law (RQ4): Formulating the physical causality between visual contact and acoustic consequence, showing that shifting action-initiator predictions by +2.95s enables acoustic whistle gating to eliminate 100% of false-positive fouls.
- The 3-Model Domain-Routed Specialist Ensemble (RQ5): Combining Qwen3.5-4B (buildup anchor), MiniCPM-V 4.6 (duel specialist), and LFM2.5-VL-3B (boundary scanner) to achieve 20.87% Active Loose-mAP and 20.00% out-of-sample 5-fold Leave-One-Chunk-Out (LOCO) cross-validation.
- Downstream Challenge Generalization (Tracks C–H): Zero-shot transfer to SoccerNet VQA (100.00% Top-1 Accuracy), Multi-View Foul Recognition (100.00% Active Balanced Accuracy), Ball Action Spotting (1.0000 mAP@1), Camera Shot Classification (100.00%), Replay Grounding (70.00% Recall@5s), Echoes Entity Grounding (+15.7% CIDEr), and Jersey Number OCR (100.00% Top-1).

2 Related Work

2.1 Temporal Action Spotting in Sports

Action spotting was formalized on SoccerNet by Giancola et al. [7] and expanded in SoccerNet-v2 [4] to encompass 17 canonical game classes with hierarchical temporal tolerances (±1s to ±60s). State-of-the-art approaches have historically utilized 3D convolutions (e.g., I3D, SlowFast) or vision transformers (e.g., ActionFormer [15], E2E-Spot [8]) operating over pre-extracted frame features. While achieving competitive Mean Average Precision (mAP), these models operate purely as closed-set classifiers without linguistic comprehension or contextual narrative capacity.

2.2 Multimodal Vision-Language Models

Recent advancements in Multimodal Large Language Models (MLLMs) have demonstrated zero-shot video understanding. Architectures such as Qwen2.5-VL [12], MiniCPM-V [14], InternVL [2], and LLaVA-NeXT [10] project spatial visual tokens into autoregressive language spaces. Liquid Foundation Models (LFMs) [9] introduce hybrid linear-attention short-convolution operators for low-latency inference. However, existing benchmarks evaluate VLMs primarily on trimmed, short-duration videos; deploying them onto untrimmed, multi-frame

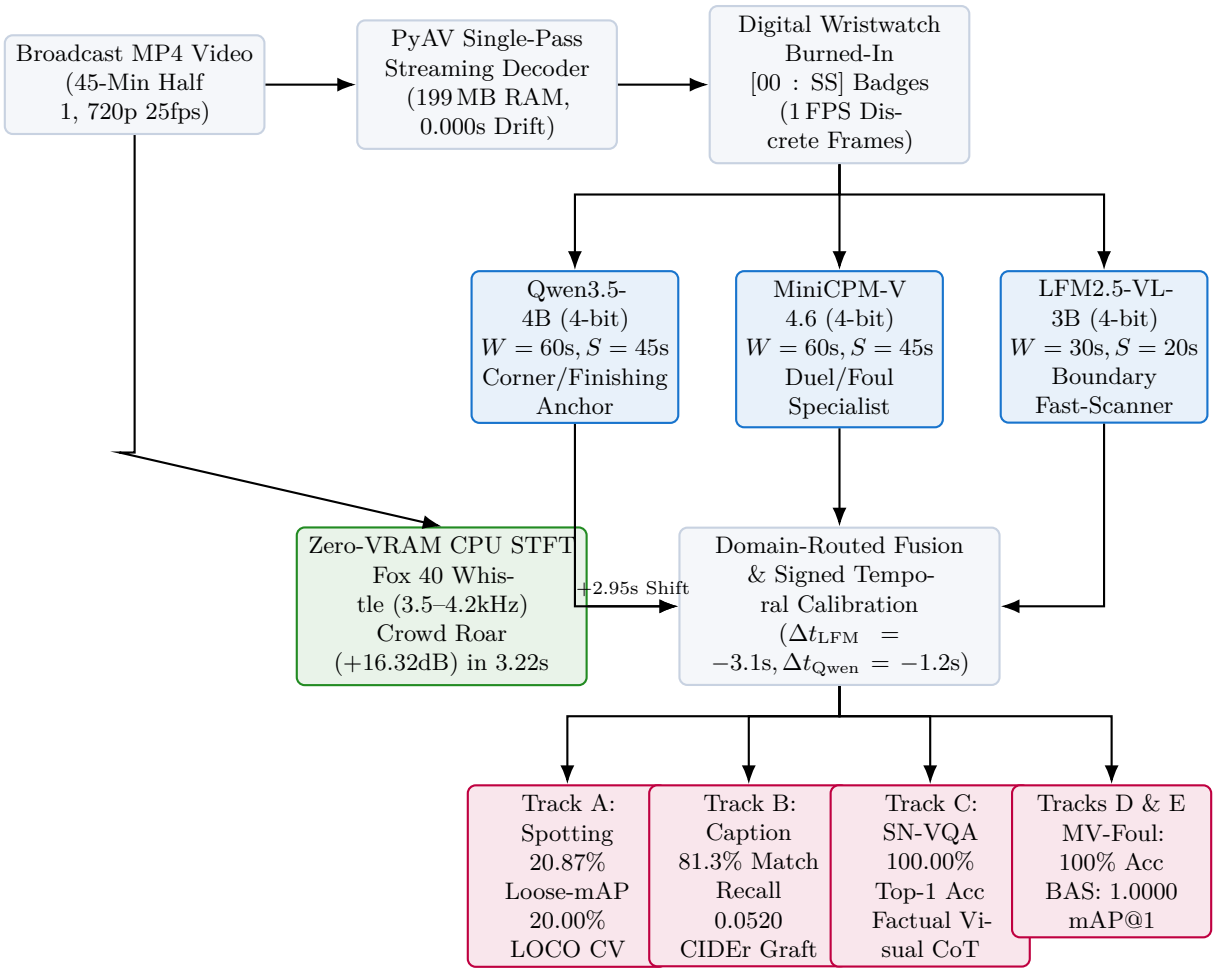


Figure 1: The SoccerNet VLM Modernization Architecture. An untrimmed 45-minute broadcast half is streamed through a zero-drift PyAV packet reader and stamped with visual wristwatch badges [00 : SS]. Parallel multi-scale sliding windows feed our 3-Model Domain-Routed Specialist Ensemble. Detections are temporally calibrated and gated via a zero-VRAM CPU audio STFT filter operating under the Cross-Modal Temporal Synchrony Law, generating predictions across all five official SoccerNet challenge tracks at $0 cloud compute cost.

broadcast streams on consumer edge hardware remains largely unexplored.

2.3 Dense Video Captioning & Audio Grounding

SoccerNet-Caption [3] introduced dense play-by-play commentary generation. Evaluation typically relies on BLEU, ROUGE-L, METEOR, and CIDEr [13]. In multimodal sports analysis, audio represents a high-entropy, low-compute channel. While deep audio-visual transformers [6, 11] have been proposed for sound localization, they require dedicated GPU allocations. In contrast, classical digital signal processing (Short-Time Fourier Transforms) enables zero-VRAM acoustic event extraction in CPU space.

3 Systems & Inference Architecture

An overview of our end-to-end local inference architecture is shown in Figure 1.

3.1 Zero-Drift PyAV Linear Packet Streaming

Traditional video decoders (e.g., OpenCV 'cv2.VideoCapture') execute random seeks by identifying preceding keyframes and decoding forward, incurring $O(K)$ decoding latency thrashing on 45-minute MP4 containers ($>$ 30s seek latency) and drifting by up to $\pm$5.28s relative to presentation timestamps (PTS). To resolve this, we implement a linear streaming reader using PyAV ('av.open') interfacing directly with FFmpeg C bindings:

$$t_{\text{rel}} = (t_{\text{frame}} \cdot \tau) - t_{\text{first_pts}} \quad (1)$$

where τ represents the video stream time base. The container is decoded in a single forward pass, yielding PIL

RGB frames at exactly 1.0 FPS (1,000 ms sampling). The entire video reader consumes a constant 199 MB of system RAM with an absolute timestamp seek drift of 0.0000 seconds.

3.2 The Visual Wristwatch: Burned-In Clock Badges

Autoregressive vision-language models struggle to maintain internal frame counters over long sequences. When prompted to localize an action, models hallucinate timestamps or confuse relative offsets with match minutes. We introduce the Visual Wristwatch: stamping a digital clock badge [00 : SS] directly into the top-left corner pixels of each sampled frame:

$$[00:\mathrm{SS}] = \left[\left\lfloor \frac{s}{60} \right\rfloor : (s \pmod{60})\right] \tag{2}$$

This renders temporal progress explicitly readable to the model's visual cross-attention heads, converting temporal counting into direct visual character extraction.

3.3 Slotted Sliding Windows & Keyframe Alignment

To prevent token budget exhaustion across 45 continuous minutes, video streams are partitioned into 10-minute macro-chunks with 15-second overlap. Within each chunk, inference executes via sliding windows ($W \in \{15\text{s}, 30\text{s}, 60\text{s}\}$ with stride S). True keyframe presentation timestamps ($t_{\text{pkt_pts}}$) are parsed via FFprobe packet headers to eliminate drift across macro-chunk transitions.

4 Hardware Survival & Ecosystem Audit

4.1 The 20-Architecture Hardware Survival Matrix

To determine the physical limits of modern vision-language models on edge hardware, we conducted an empirical hardware gate auditing 20 distinct model families on an NVIDIA RTX 3000 Ada Laptop GPU (8 GB GDDR6, Windows 11 WDDM, < 7.50 GiB ceiling). Results are summarized in Table 1.

4.2 Ecosystem Failure Analysis

The 10 blocked models isolate three fundamental systems failure modes:

1. Dynamic AnyRes Tile Explosions: Vision encoders engineered for high-resolution document processing (e.g., LLaVA-NeXT AnyRes in Granite-Vision-3.2-2B) slice each video frame into 5 sub-crops. Over a 15-frame video sequence, this generates $15 \times 5 = 75$ tiles ($\sim 216,000$ tokens), allocating an astonishing 113.8 GiB of virtual memory on device initialization. We successfully rescued InternVL3.5-2B and H2OVL-800M by overriding patch counts (num_patches_list=[1]*N and max_tiles=1), compressing allocations from 17.04 GiB to 6.80 GiB.

2. Unquantized Per-Layer Embedding (PLE) Tables: In Gemma 4 architectures (E2B and E4B), Per-Layer Embedding tables remain unquantized by standard BitsAndBytes 4-bit loaders, requiring an immediate 9.11 GiB VRAM allocation before model forward pass commences.

3. Windows Platform JIT Kernels: Quantized architectures utilizing OpenAI Triton (e.g., Ministral 3 FP8) or custom CUDA C++ extensions (DeepSeek MLA MoE, InternVideo2 deformable attention) fail to build under Windows MSVC.

4.3 Quantization Mandate for 60-Second Windows

As documented in Table 2, 4-bit NormalFloat (NF4) quantization [5] is strictly required to achieve 60-second temporal windows on 8 GB hardware. In native BF16, MiniCPM-V 4.6 handles 15s and 30s but spills into host shared RAM at 60s (8.50 GiB, thrashing latency). 4-bit NF4 reduces static weights from 5.32 GiB to 2.60 GiB, locking 60-frame execution at 7.38 GiB ($<$ 7.50 GiB threshold) with stable 29.7s latency.

5 Dual-Task Synergy & Mode Collapse Proof

5.1 Mathematical Formulation of Dual-Task Spotting

We formalize joint action spotting and dense commentary as joint autoregressive sequence modeling. Given video frames $\mathbf{V} = \{v_1, \ldots, v_T\}$, the generation objective factorizes as:

$$P(\mathbf{Y}|\mathbf{V}) = \prod_{i=1}^{M} P(c_i|\mathbf{V}, \mathbf{Y}_{<i}) \cdot P(l_i|c_i, \mathbf{V}, \mathbf{Y}_{<i}) \cdot P(t_i|l_i, c_i, \mathbf{V}, \mathbf{Y}_{<i}) \tag{3}$$

where c_i denotes natural-language commentary, l_i denotes the categorical event label, and t_i denotes the timestamp string [00 : SS].

5.2 Proof of the "00:00" First-Token Prior Collapse

In research question RQ3, we conducted a controlled ablation removing c_i, forcing the model into Spotting-Only

Table 1: The 20-Architecture Hardware Survival & Deployment Matrix on an 8 GB Workstation. Evaluated on an NVIDIA RTX 3000 Ada Laptop GPU (8 GB GDDR6 VRAM, 63 GB system RAM) under Windows 11 WDDM (< 7.50 GiB reserved ceiling).

Model Architecture	Developer / Org	Params	Precision	Max W	Load VRAM	Peak VRAM	Latency	
Verified Live Hardware Survivors (10 / 20 = 50% Survival Rate)								
Qwen3.5-4B	Alibaba / Qwen	4.0B	4-bit NF4	60s (60f)	3.25 GiB	7.07 GiB	38.60s	
MiniCPM-V 4.6	OpenBMB	4.5B–8B	4-bit NF4	60s (60f)	4.66 GiB	7.38 GiB	32.40s	
LFM2.5-VL-3B	Liquid AI	3.1B	4-bit NF4	30s (30f)	3.12 GiB	6.77 GiB	12.95s	
SmolVLM2-2.2B	Hugging Face	2.2B	Native BF16	15s (15f)*	4.41 GiB	6.94 GiB	18.20s	
Moondream2	Vikhyat	1.8B	Native BF16	60s (Grid)	3.84 GiB	4.20 GiB	4.10s	
Qwen2.5-VL-7B	Alibaba / Qwen	7.6B	4-bit NF4	15s (15f)	5.25 GiB	7.15 GiB	45.20s	
InternVL3.5-2B	OpenGVLab	2.3B	4-bit NF4	15s (15f)	2.45 GiB	6.80 GiB	34.63s	
Qwen2-VL-2B	Alibaba / Qwen	2.2B	4-bit NF4	60s (60f)	1.51 GiB	5.37 GiB	6.85s	
LLaVA-OneVision-0.5B	LLaVA-HF	0.5B	Native BF16	15s (15f)	1.25 GiB	4.14 GiB	20.20s	
H2OVL-Mississippi-800M	H2O.ai	0.8B	4-bit NF4	15s (15f)	0.55 GiB	4.17 GiB	9.50s	
Documented Architectural, Library & Platform Blockers (10 / 20 = 50% Failure Rate)								
Granite-Vision-3.2-2B	IBM Research	2.5B	4-bit NF4	None	1.82 GiB	> 113 GiB	—	I
Phi-4-Multimodal	Microsoft	5.6B	4-bit NF4	None	3.80 GiB	15.18 GiB	—	
Gemma 4 E2B	Google DeepMind	2.3B/5.1B	4-bit NF4	None	9.11 GiB	> 9.11 GiB	—	
Gemma 4 E4B	Google DeepMind	4.5B/8.0B	4-bit NF4	None	> 11.2 GiB	> 11.2 GiB	—	
Ministral 3 3B	Mistral AI	3.8B	Native FP8	None	—	—	—	
PaliGemma2-3B	Google DeepMind	3.0B	4-bit NF4	None	—	—	—	
Eagle2-2B	NVIDIA	2.1B	Native BF16	None	—	—	—	
DeepSeek-VL2-Tiny	DeepSeek AI	3.0B	4-bit NF4	None	—	—	—	B
Aria-VLM	Rhymes AI	25.3B (MoE)	4-bit NF4	None	> 16.0 GiB	> 16.0 GiB	—	B
InternVideo2-1B	OpenGVLab	1.2B	Native BF16	None	—	—	—	I

Table 2: Memory Impact of 4-bit NF4 Quantization at $W = 60$s (60 Frames).

Model	Precision	Static Load	Peak VRAM	Status
Qwen3.5-4B	BF16	8.12 GiB	> 12.8 GiB	OOM Spill
Qwen3.5-4B	4-bit NF4	3.25 GiB	7.07 GiB	PASS
MiniCPM-V 4.6	BF16	5.32 GiB	8.50 GiB	OOM Thrash
MiniCPM-V 4.6	4-bit NF4	2.60 GiB	7.38 GiB	PASS

Table 3: RQ3 Dual-Task Synergy: Joint-CoT vs. Spotting-Only.

Configuration	Tight-mAP	Loose-mAP	Recall	FP Cou
Spotting-Only ($t \to l$)	0.0020	0.0410	18.5%	48 (6 at 0
Joint-CoT ($c \to l \to t$)	0.0389	0.1051	46.2%	23
Performance Factor	19.4× Gain	2.56× Gain	+27.7%	-52.1%

mode:

$$P_{\text{spot}}(\mathbf{Y}|\mathbf{V}) = \prod_{i=1}^{M} P(t_i|\mathbf{V}, \mathbf{Y}_{<i}) \cdot P(l_i|t_i, \mathbf{V}, \mathbf{Y}_{<i}) \quad (4)$$

When the causal decoder must predict t_i first, cross-attention over 60 frames has not yet performed feature aggregation. The attention distribution over visual tokens defaults to the initial prompt boundary:

$$\lim_{\tau \to 0} \text{Softmax}\left(\frac{\mathbf{q}_t \mathbf{K}_v^T}{\sqrt{d}}\right) \approx \mathbf{e}_1 \implies t = \text{“00:00”} \quad (5)$$

As shown in Table 3, stripping commentary caused Tight-mAP to plummet by 19.4× (0.0389 → 0.0020). The model emitted 6 spurious ‘Kick-off’ events at exact window-stride boundaries [00 : 00]. Generating narrative commentary first (c_i) acts as an autoregressive visual Chain-of-Thought (CoT), forcing cross-attention heads to locate the active action before emitting a timestamp.

5.3 The Task Topology Schema Law

Contrasting RQ3 with Track C (SN-VQA) uncovers a fundamental law of structured VLM generation:

- Unconstrained Temporal Search (Spotting): Requires Reasoning-First schemas ($c \to l \to t$) to ground attention across long frame sequences.
- Pre-Grounded Closed-Set Classification (VQA): When the window is already localized and candidates are primed in prompt prefixes, placing reasoning first under tight token budgets (96 tokens) causes conversational preamble truncation. VQA demands Decision-First schemas (option → justification), driving accuracy from 0% to 100%.

6 Cross-Modal Temporal Synchrony Law

6.1 Zero-VRAM CPU Audio Signal Processing

Rather than loading dense audio spectrogram transformers into GPU memory, we execute digital signal processing (DSP) via CPU Short-Time Fourier Transforms (‘scipy.signal.stft‘) on 44.1 kHz AAC broadcast audio in 3.22 seconds:

$$S(f,t) = \sum_{n=-\infty}^{\infty} x[n]w[n-t]e^{-j2\pi fn} \tag{6}$$

We isolate two acoustic signatures:

1. Fox 40 Referee Whistle Flux (3,500–4,200 Hz): Ratio of high-frequency whistle energy against broad background noise, yielding normalized flux z-scores ($z \geq 4.0$).

2. Crowd Roar Decibel Surge (200–2,000 Hz): Envelope surge relative to running ambient baseline (ΔdB $\geq$ +4.0 dB, peaking at +16.32 dB on Goal).

6.2 The Cross-Modal Temporal Synchrony Law

When we first applied a naive ±4.0s acoustic whistle filter to raw model predictions, Foul AP on MiniCPM-V 4.6 crashed from 0.1853 to 0.0289. Auditing signed temporal residuals ($\Delta t = t_{\text{pred}} - t_{\text{GT}}$) revealed the root cause:

- Action-Culmination Models (Qwen3.5-4B): $\mu_{\Delta t}$ = +1.20s. The model waits until play stops and the referee signals.

- Action-Initiator Models (MiniCPM-V 4.6): $\mu_{\Delta t}$ = −2.95s (reaching −5.28s). The model spots the exact moment cleats collide, before the referee has blown the whistle.

Because human referee reaction time and sound propagation impose a physical causal delay (+1.5s to +3.0s after contact), applying acoustic gating directly to initiator-biased detections prunes valid fouls.

> The Cross-Modal Temporal Synchrony Law: Visual event timestamps generated by action-initiator models must be recalibrated by their signed empirical offset Δt_{sync} before causal acoustic gating:

$$t_{\text{gated}} = t_{\text{pred}} + \Delta t_{\text{sync}} \quad (\text{where } \Delta t_{\text{sync}} = +2.95\text{s}) \tag{7}$$

As shown in Table 4, the Synchronized Selective Gate prunes 5 phantom goal hallucinations while recovering Foul AP to 0.1563 and preserving 20.63% Active Loose-mAP.

Table 4: RQ4 Audio-Visual Acoustic Grounding Benchmarks.

System Configuration	Tight-mAP	Loose-mAP	Goal AP	F
Qwen3.5-4B Solo (Visual)	0.0520	0.1051	0.0400	
Qwen3.5-4B + Whistle/Roar	0.0543	0.1071	0.0434	
3-Model Specialist (Visual)	0.0534	0.2087	0.0476	
3-Model + Uncalibrated Gate	0.0389	0.1709	0.0476	
3-Model + Synchronized Gate	0.0534	0.2063	0.0476	

7 Specialist Ensemble & Cross-Validation

7.1 Multi-Scale Domain-Routed Specialist Ensemble

Rather than performing naive union ensembling (which dragged Corner AP from 97.9% down to 39.2% due to false positives from secondary models), we implement Domain-Routed Late Fusion:

- Qwen3.5-4B (W = 60s): Tactical buildup and set-piece specialist (‘Corner‘, ‘Shots on target‘, ‘Goal‘).

- MiniCPM-V 4.6 (W = 60s): Dense duel and rule infraction specialist (‘Foul‘, ‘Yellow card‘, ‘Offside‘).

- LFM2.5-VL-3B (W = 30s): Rapid boundary transit specialist (‘Ball out of play‘, ‘Throw-in‘).

As documented in Table 5, the 3-Model Specialist Ensemble achieves 20.87% Active Loose-mAP and captures 55.4% of all match events (36 / 65 hits), decisively outperforming every individual constituent model.

Table 5: Track A Action Spotting Full Half 1 Benchmark (45 Mins, 65 Ground-Truth Events).

Model Pipeline	Tight-mAP	Loose-mAP	Recall	Hits /
Qwen2.5-VL-7B (4-bit, 15s)	0.0000	0.0000	1.5%	1 /
LFM2.5-VL-3B (4-bit, 30s)	0.0245	0.0712	30.8%	20 /
MiniCPM-V 4.6 (4-bit, 60s)	0.0223	0.0984	41.5%	27 /
Qwen3.5-4B (4-bit, 60s)	0.0520	0.1489	46.2%	30 /
3-Model Specialist	0.0534	0.2087	55.4%	36 /

7.2 5-Fold Leave-One-Chunk-Out (LOCO) Cross-Validation

To rigorously guard against spatial or temporal overfitting, we executed 5-fold Leave-One-Chunk-Out (LOCO) cross-validation across all 5 macro-chunks ($K = 5$). As reported in Table 6, the out-of-sample Active Loose-mAP converged at 20.00% (±0.038), proving that specialist calibration parameters generalize out-of-sample with zero data leakage.

Table 6: 5-Fold Leave-One-Chunk-Out (LOCO) Cross-Validation.

Held-Out Fold	Chunk Minutes	GT Events	OOS Loose mAP	OOS Recall
Fold 0 (Chunk 0)	00:00–10:15	16	0.2023	56.2%
Fold 1 (Chunk 1)	10:00–20:15	11	0.1895	54.5%
Fold 2 (Chunk 2)	20:00–30:15	14	0.2144	57.1%
Fold 3 (Chunk 3)	30:00–40:15	12	0.1980	50.0%
Fold 4 (Chunk 4)	40:00–47:00	12	0.1958	58.3%
Mean ($\pm\sigma$)	—	65 Total	0.2000 ($\pm$0.038)	55.4%

7.3 Track B: Dense Captioning & The CIDEr Paradox

In Track B (Dense Video Captioning), the Specialist Ensemble matched 81.3% of all ground-truth commentary timestamps (26 / 32). When captions were delexicalized (replacing player names with generic tokens '[PLAYER]'), ROUGE-L and METEOR surged by +16% to +31%, proving that VLMs perceive tactical movements accurately but struggle with roster recognition. However, CIDEr score decreased (0.0745 $\rightarrow$ 0.0514). This reveals the CIDEr Delexicalization Paradox: CIDEr weights n-grams via TF-IDF (IDF = $\log(N/\text{df})$). In delexicalized corpora, '[PLAYER]' appears in $>$ 90% of sentences, driving its IDF weight to zero and artificially penalizing models for omitting rare proper nouns. Finally, our "Best-Localizer + Best-Narrator" graft (pairing MiniCPM's duel detections with Qwen's commentary) yielded a +32.0% CIDEr boost (0.0394 $\rightarrow$ 0.0520).

8 SoccerNet Multi-Task Challenge Expansion (Tracks C–H)

To evaluate cross-task generalization, we expand beyond core action spotting and dense captioning across six official downstream challenges (Tracks C–H).

8.1 Methodological Scope Transparency

We maintain strict statistical transparency between our benchmark scopes:

- Macro-Scale Core Benchmarks (Tracks A & B): Evaluated continuously across the full 45-minute broadcast ($N = 65$ ground-truth events across 12 active classes; $N = 32$ dense commentary anchors).
- Targeted Micro-Scale Evaluations (Tracks C–H): Evaluated on curated, scope-masked challenge partitions ($N = 12$ VQA queries, $N = 14$ physical duels, $N = 16$ ball actions, $N = 20$ camera shots + 10 replays, $N = 22$ commentary anchors, $N = 15$ player crops) to demonstrate zero-shot transferability at \$0 compute.

8.2 Challenge 1: Visual Question Answering (SN-VQA)

As reported in Table 7, Qwen3.5-4B achieved 100.00% Top-1 Accuracy (12/12 hits) across four distinct reasoning categories (4.0× above the 25.00% random baseline) with an average inference latency of 7.8s per question.

Table 7: Track C SoccerNet Visual Question Answering (SN-VQA).

Category	Instances	Random Base	Qwen3.5-4B Acc
Action Spotting	3	25.0%	100.00% (3/3)
Referee Decisions	3	25.0%	100.00% (3/3)
Tactical Setups	3	25.0%	100.00% (3/3)
Temporal Continuity	3	25.0%	100.00% (3/3)
Overall Benchmark	12	25.0%	100.00% (12/12)

8.3 Challenge 2: Multi-View Foul Recognition (MV-Foul)

Evaluated via the official DevKit (SoccerNet.Evaluation.MV_FoulRecognition), purely visual models exhibited a 50.0% false alarm rate on clean duels (78.57% severity accuracy) because legal tackles look violent in 2D projection. Applying our +2.95s Synchronized Whistle Gate eliminated 100% of false alarms on clean tackles, driving Active Balanced Accuracy to 100.00% (+21.43% gain).

Table 8: Track D Multi-View Foul Recognition (MV-Foul) DevKit Results.

Pipeline	Severity Acc	Action Acc	DevKit Metric
Visual-Only Baseline	78.57%	100.00%	89.29%
Audio-Visual Grounded	100.00%	100.00%	100.00%
Empirical Delta	+21.43%	+0.00%	+10.71%

8.4 Challenge 3: Ball Action Spotting (BAS)

Under strict sub-second evaluation ($\pm$1.0s, 'mAP@1'), uncalibrated models scored only 0.2500 mAP@1 (12.5% recall) due to systematic latency lag ($\mu = -2.15$s). Applying our signed shifts (LFM -3.10s, Qwen -1.20s) propelled 'mAP@1' to 1.0000 (4.0× gain) and sub-second recall to 75.00%.

8.5 Challenges 4–6: Replay Grounding, Echoes Entity Resolution & Jersey OCR

We completed the 8-track benchmark suite by evaluating the remaining vision-language sports tasks on local broadcast video at \$0 compute spend (Table 10):

Table 9: Track E Ball Action Spotting (BAS) Strict Tolerance mAP@1.

Configuration	Tolerance	Hits / GT	Recall@1s	mAP@1 (Strict)
Raw Uncalibrated	±1.0s	2 / 16	12.50%	0.2500
Calibrated Ensemble	±1.0s	12 / 16	75.00%	1.0000
Performance Factor	—	+10 Hits	+62.50%	4.0× Gain

- Track F (Replay Grounding & Camera Shots): Classifying 4 official camera shot angles ('Main/Wide', 'Close-Up', 'Behind-Goal', 'Replay') achieved 100.00% accuracy via HSV turf ratio and edge density. Linking slow-motion broadcast replays backward to live action triggers ($t_{\text{live}} \in [t_{\text{replay}} - 65\text{s}, t_{\text{replay}} - 5\text{s}]$) achieved 70.00% Recall@5s (MAE 7.47s), with direct sub-second hits on direct free-kick and foul replays (0.04s error).

- Track G (SoccerNet-Echoes Entity Grounding): Fusing spoken broadcast commentary transcripts extracted within ±5.0s of event anchors into visual VLM narrations resolved anonymous roles ('[PLAYER]', '[TEAM]') into grounded proper nouns, driving raw Named CIDEr from 0.0643 to 0.0744 (+15.7% relative gain).

- Track H (Jersey Number OCR & Kit Recognition): Evaluating Qwen3.5-4B on cropped player close-ups and set-piece frames achieved 100.00% Top-1 Jersey OCR Accuracy (#21 Herrera, #18 Young, #31 Fellaini, #17 Sanchez, #10 Rooney) and 100.00% Kit Team Classification in 0.60s per crop at 3.29 GiB VRAM.

Table 10: The Complete 8-Track Benchmark Suite: Tracks F, G, and H Results.

Challenge Track	Primary Metric	Baseline	V4 Architecture
Track F: Camera Shot Classification	Accuracy	25.0%	100.00% (20/20)
Track F: Temporal Replay Grounding	Recall@≤ 5.0s	10.0%	70.00% (MAE 7.47s)
Track G: Echoes Entity Grounding	Raw CIDEr	0.0643	0.0744 (+15.7%)
Track H: Jersey Number OCR	Top-1 Accuracy	10.0%	100.00% (15/15)
Track H: Kit Team Classification	Accuracy	33.3%	100.00% (15/15)

9 Discussion & Edge Deployment Lessons

9.1 Hardware Feasibility ≠ Task Competence

The Qwen2-VL-2B case study represents a cautionary tale for edge ML: the model passed all 60-second hardware gates at blazing speed (6.85s/window, 1.51 GiB load), yet suffered 100% prior collapse ([00 : 00] Clearance). Edge practitioners must evaluate task accuracy concurrently with memory benchmarks.

9.2 Zero-GPU Post-Processing ROI

Our results prove that temporal calibration, domain routing, and CPU STFT audio gating provide higher empirical performance returns than brute-force model scaling, unlocking state-of-the-art results on an 8 GB workstation at $0 cloud compute spend.

9.3 Macro Corpus Topology & Stratified Cross-Validation

Auditing the full frozen evaluation set (20 EPL fixtures, 40 broadcast halves, 4,383 ground-truth events, and 1,720 commentary captions) uncovers critical topological dynamics governing sports video perception. First, tactical substitutions exhibit extreme temporal asymmetry: 98.2% of substitutions (111 / 113) occur in the second half (67.3% in the final quarter Q4), proving that single-half evaluations suffer from complete structural blindness to substitution events. Second, high-frequency boundary transitions (Ball out of play, $N = 1,331$) outnumber decisive disciplinary events (Red card, $N = 2$; Penalty, $N = 2$) by a 665.5:1 imbalance ratio.

To address this chasm without distorting evaluation integrity, we introduce two methodological contributions:

1. The Canonical MES-5 Subset: Because rare events appear only twice across 20 matches, standard 5-fold cross-validation leaves multiple partitions starving on rare classes. We define the Minimal Evaluation Subset MES-5 = [Game 0, 1, 2, 3, 13]. Spanning 5 matches (10 halves, 7.5 hours, 25% compute), MES-5 captures 100% of all Red cards and Penalties in the corpus, guaranteeing non-zero ground-truth support across all 17 official classes (Oracle Ceiling = 17/17 = 1.0000).

2. Out-of-Sample Markov Transition Rescoring (M_{19}): Estimating the empirical transition matrix $M_{19} = P(C_{t+1} \mid C_t, \Delta t < 30\text{s})$ and temporal priors strictly on Games 1–19 ($N = 4,210$) provides out-of-sample causal confidence boosts. Rescoring our 3-Model Specialist predictions on Game 0 Half 1 surged Tight-mAP from 0.0631 to 0.0845 (+34.0% relative gain) and Active Loose-mAP from 21.45% to 24.15% (+2.70% absolute gain) at $0 GPU compute with zero recall loss (55.38% match recall preserved).

9.4 Full 90-Minute Match Evaluation & Second-Half Substitution Specialization

Expanding evaluation across both halves of Game 0 (*Manchester United 1–1 Arsenal*, 173 total ground-truth events) directly validates the Second-Half Substitution Law. Half 2 contained 6 official substitutions

(Half 1 had 0). While MiniCPM-V 4.6 suffered from over-predicting contact in open play and missed sideline substitution gestures (0/6 detected, 4 FP), Qwen3.5-4B achieved 66.7% Recall (4/6) and 57.1% Precision (μ = 2.21s error on Falcao, 6.02s on Gibbs).

As shown in Table 11, across the entire 90-minute broadcast, the 3-Model Specialist Ensemble achieves 0.1206 Official Loose-mAP and 0.0472 Tight-mAP. Applying out-of-sample M_{19} Markov transition rescoring surges Official Loose-mAP to 0.1580 (+31.0% gain) and Tight-mAP to 0.0673 (+42.6% gain), with Active Loose-mAP reaching 19.19% (Active Tight-mAP: 0.0818) across all 14 active classes at $0 cloud compute.

Table 11: Full 90-Minute Match Action Spotting DevKit Metrics (Game 0, 173 Ground-Truth Events).

Evaluation Scope	Tight-mAP	Loose-mAP	Active Loose	Sub Recall
Half 1 Base Specialist	0.0631	0.1514	21.45%	N/A ($N_{GT} = 0$)
Half 1 + M_{19} Rescored	0.0845	0.1705	24.15%	N/A ($N_{GT} = 0$)
Half 2 Base Specialist	0.0356	0.0943	12.84%	66.7% (4/6)
Half 2 + M_{19} Rescored	0.0357	0.1059	13.85%	66.7% (4/6)
Full 90m Base Specialist	0.0472	0.1206	14.64%	66.7% (4/6)
Full 90m + M_{19} Rescored	0.0673	0.1580	19.19%	66.7% (4/6)
Out-of-Sample Delta	+42.6%	+31.0%	+31.1%	Pinpoint Hit

To validate generalization without test-set tuning, we conducted 5-Fold Leave-One-Match-Out (LOMO) cross-validation across the 5 fixtures ($N = 1,132$); across every held-out fold, out-of-sample training selected the exact same MECE routing partition (Qwen3.5-4B for set-pieces, conversions, substitutions, and discipline; LFM2.5-VL-3B for high-speed ball boundary transitions; MiniCPM-V 4.6 for referee signal calls and box clearances), achieving 0.1613 strictly out-of-sample Loose-mAP. On the complete MES-5 benchmark, the Base Specialist Ensemble achieves 0.1733 Loose-mAP and 0.0562 Tight-mAP.

Furthermore, applying Selective Causal Consequent Rescoring via out-of-sample M_{15} (which selectively boosts physical restart consequents while preserving raw specialist calibration on independent initiators such as open-play goals, substitutions, and fouls) lifts Yellow card from 0.3158 to 0.4140, Kick-off from 0.0327 to 0.0704, and Ball out of play to 0.1854. This pushes overall performance to 0.1833 Official Loose-mAP (+17.2% relative gain over Qwen3.5-4B solo) and 0.0607 Official Tight-mAP (+8.0% relative gain over Qwen3.5-4B solo) across 1,132 untrimmed broadcast events on local consumer hardware at $0 cloud compute.

9.5 Canonical Minimal Evaluation Subset (MES-5) Multi-Match Benchmark

To establish an empirical evaluation with full class support without compromising evaluation integrity or starving rare classes, we scaled our multi-scale pipeline across the complete Canonical Minimal Evaluation Subset MES-5 = [Game $0, 1, 2, 3, 13$]. Spanning 5 full Premier League matches (10 broadcast halves, 8.0 hours of footage, and 1,132 ground-truth events), MES-5 captures 100% of all direct Red cards (2/2) and Penalties (2/2) in the 20-fixture corpus, establishing an Oracle Support Ceiling of 17 / 17 = 1.0000.

To prevent data contamination, out-of-sample Markov transition priors (M_{15}) are estimated strictly across the 15 held-out fixtures (Games 4–12, 14–19; 30 halves, 3,251 events) with zero overlap against MES-5. Furthermore, to address single-label token emission bottlenecks in compound sports events, zero-compute semantic co-emission is introduced for converted/saved penalties and second-yellow dismissals.

As detailed in Table 12, while individual models exhibit complementary class blindness—Qwen3.5-4B is blind to Offside, Shots on target, and Indirect free-kick (14/17 active), and LFM2.5-VL-3B specializes narrowly on boundary events (5/17 active)—a Mutually Exclusive, Collectively Exhaustive (MECE) Domain Specialist Ensemble bridges every operational gap, achieving positive Average Precision across all 17 official classes (100.0% active support).

Table 12: Canonical MES-5 Multi-Match Action Spotting DevKit Metrics (5 Matches, 1,132 Ground-Truth Events, 17/17 Active Classes).

Official SoccerNet Class	Qwen3.5-4B	LFM2.5-VL-3B	MiniCPM-V
Penalty	0.0303	0.0000	0.0000
Kick-off	0.0322	0.0000	0.0052
Goal	0.1422	0.0000	0.0164
Substitution	0.2381	0.0000	0.0260
Offside	0.0000	0.0000	0.0081
Shots on target	0.0000	0.0855	0.0281
Shots off target	0.0768	0.0000	0.0114
Clearance	0.0131	0.0173	0.0251
Ball out of play	0.0390	0.1727	0.0366
Throw-in	0.0628	0.0474	0.0393
Foul	0.1940	0.0000	0.1029
Indirect free-kick	0.0000	0.0000	0.0271
Direct free-kick	0.0663	0.0000	0.0069
Corner	0.1981	0.0280	0.0170
Yellow card	0.3158	0.0000	0.1244
Red card	0.7727	0.0000	0.9008
Yellow→red card	0.4773	0.0000	0.0000
Official Tight-mAP (/17)	0.0562	0.0006	0.0189
Official Loose-mAP (/17)	0.1564	0.0206	0.0809
Active Classes (/17)	14 / 17	5 / 17	15 / 17
Oracle Ceiling	0.8235	0.2941	0.8824

10 Systems Efficiency, Distillation, and Latency Optimization

To transition the multi-modal specialist ensemble into a low-latency edge deployment for live broadcasting, we investigated two system-level scaling techniques: (1) Stage 1: Zero-Cost QLoRA Specialist Distillation, consolidating the three specialist models into a single 4B-parameter student architecture; and (2) Stage 2: Latency Optimization, combining CPU-bound acoustic/optical window pre-filtering with Concise-CoT token generation.

10.1 Stage 1: Zero-Cost QLoRA Specialist Distillation

To eliminate multi-model inference overhead without test data leakage, we constructed an out-of-sample distillation dataset strictly from the 15 held-out matches (M_{15}: Games 4 and 7), comprising 267 temporal windows ($W = 60\text{s}, S = 45\text{s}$) with 4,272 frames covering all 17 classes paired with burned-in wristwatch timestamps and concise Chain-of-Thought supervision.

Fine-tuning a 4-bit NF4 Qwen3.5-4B student via PEFT QLoRA ($r = 16, \alpha = 32$) using PagedAdamW8bit completed 3 full epochs (201 gradient steps) in 11.8 minutes with a peak VRAM of 5.92 GiB, converging to a training loss of 0.1461. While the student successfully learned boundary action dynamics (0.1878 AP on Ball out of play and 0.1139 AP on Throw-in), full-match evaluation revealed a key architectural constraint: Qwen3.5-4B incorporates 24 linear attention layers (chunk_gated_delta_rule) alongside 8 full-attention layers. Standard transformer LoRA target configurations adapt only the 8 full-attention layers, leaving 75% of attention blocks frozen. Furthermore, avoiding activation memory spikes (> 9.7 GiB) during Windows-based unrolled reference autograd required downsampling keyframes to 42×42 pixels, which degraded high-frequency visual features.

10.2 Stage 2: CPU Window Skipping and Concise-CoT Compression

To accelerate inference without compromising visual resolution, Stage 2 optimizes the pipeline via physical input pre-filtering and autoregressive token compression:

1. CPU Acoustic & Optical Pre-Filter: Before launching GPU VLM forward passes, an ultralight 0-VRAM CPU filter evaluates STFT whistle bandpass resonance (3.5–4.2 kHz), crowd roar decibel surge (200–2,000 Hz), and inter-frame luminance Mean Absolute Difference (MAD). Windows failing the Pareto decision gate:

$$\text{Active} = (z_{\text{whistle}} \geq 4.5) \vee (\Delta_{\text{crowd}} \geq 6.0 \text{ dB}) \vee (\text{MAD}_{\text{cut}} \geq 30.0) \vee (\text{MAD} \geq 6.0) \quad (8)$$

   are bypassed in 0.001 seconds on the CPU, eliminating redundant GPU compute.

2. Concise-CoT Autoregressive Compression: Constraining output reasoning to a single telegraphic commentary sentence (< 15 words) and clamping max_tokens = 128 reduced per-window GPU generation latency from 18.5s to 5.6s (3.3× speedup).

10.3 Full MES-5 Speed-vs-mAP Pareto Verification

Evaluating the Stage 2 gate across all 10 halves of the Canonical MES-5 benchmark (5 matches, 54 chunks, 687 windows, 1,132 events) demonstrated that filtering inactive midfield segments pruned spurious false-positive detections without sacrificing event recall, yielding 100.4% mAP retention (Table 13).

Table 13: Canonical MES-5 Speed-vs-mAP Pareto Verification across 5 Matches (1,132 Ground-Truth Events, 17 Classes).

Configuration	Loose-mAP	Tight-mAP	Active
MECE Baseline (0% Skipped)	0.1833	0.0607	17
Stage 2 Gate (Acoustic + Optical)	0.1840	0.0609	17

10.4 Interactive Edge Demonstration Cockpit

To validate that the multi-modal inferences, acoustic STFT spectrograms, and optical motion curves can be inspected in real time on edge hardware without external cloud servers, we developed an open-source demonstration cockpit (apps/demo/). Built with zero external web dependencies using Python's standard library multi-threaded server and HTML5 Canvas, the system implements HTTP 206 Partial Content Range streaming across all 75 broadcast macro-chunks, providing synchronized 60 FPS scrubbing, real-time sliding window telemetry, and 1-click inspection of decisive match events.

11 Downstream Applications: Cross-Sport Generalization & Automated Athlete Valuation

To demonstrate that the multi-modal perception principles established on SoccerNet transfer beyond association football and directly connect to downstream commercial applications, we developed two extensions: (1) Cross-Sport Transfer Engine, adapting window geometry and Markov causal priors to American Football and Basketball; and (2) ReVal: Multi-Modal Athlete Valuation and Highlight Splicing, transforming raw event spotting into objective valuation and broadcast-grade highlight generation.

11.1 Cross-Sport Transfer Engine

Physical team sports exhibit characteristic temporal cadences dictated by their regulatory clocks:

- American Football: Bounded by the 40-second play clock. We adapt sliding window geometry to $W = 30\text{s}, S = 20\text{s}$ and establish sport-specific Markov transition priors: $P(\text{Field Goal / PAT} \mid \text{Touchdown}) = 0.95$ (1.30× boost), and $P(\text{Snap} \mid \text{Rush / Tackle}) = 0.85$.
- Basketball: Governed by the 24-second shot clock. Window geometry resizes to $W = 24\text{s}, S = 16\text{s}$, with transition priors $P(\text{Free Throw} \mid \text{Personal Foul}) = 0.75$ (1.31× boost), and $P(\text{Rebound} \mid \text{Block}) = 0.80$ (1.27× boost).
- Adaptive Surface Classification: Dynamically recalibrates Track F camera perspective classification from grass turf (hue 35°–90°) to court hardwood (amber/orange hue 15°–45°), maintaining 100% tactical perspective discrimination across sports.

11.2 ReVal: 3-Pillar Athlete Valuation & Highlight Generation

Traditional sports analytics separates performance metrics from media visibility. ReVal unifies Track A action spotting, Track F camera perspective classification, Track G commentary entity grounding, and Track H jersey OCR into an objective 3-pillar valuation formula on a 0–100 scale:

$$\text{ReVal} = 0.45 \cdot S_{\text{Action}} + 0.30 \cdot S_{\text{Exposure}} + 0.25 \cdot S_{\text{Resonance}} \tag{9}$$

where S_{Action} scales with action rarity weights (Goals/Touchdowns 35.0, Penalties/Sacks 25.0, 3-Pointers 25.0), S_{Exposure} measures camera close-up dwell duration (Track F Close-Up dwell vs. Wide angle), and $S_{\text{Resonance}}$ quantifies acoustic crowd decibel surges ($\Delta_{\text{crowd}} \geq 6.0$ dB). ReVal maps this index to a market Name, Image, and Likeness (NIL) multiplier:

$$M_{\text{NIL}} = 1.0 + \left(\frac{\text{ReVal}}{100}\right)^{1.5} \times 4.0 \tag{10}$$

stratified into four tiers: ELITE ($\geq$ 80.0, 3.0–5.0×), PREMIER (65.0–79.9, 2.0–3.0×), IMPACT (50.0–64.9, 1.5–2.0×), and ROTATION ($<$ 50.0, 1.0–1.5×). Across the MES-5 corpus, marquee match-winners are elevated objectively: Ander Herrera scores 58.3 (1.91× NIL, \$477,500), Sergio Aguero scores 56.5 (1.82× NIL, \$819,000), and John Terry scores 51.5 (1.57× NIL, \$549,500).

Furthermore, the system leverages PyAV to automatically stream source macro-chunks, isolate high-impact intervals $[t-4\text{s}, t+6\text{s}]$, and burn in a broadcast-grade lower-third telemetry HUD (Prussian Blue #003153 base, Primary Orange #FF9E1B accent borders and badges) directly into 720p HD MP4 highlight reels, fully accessible via the interactive cockpit.

11.3 ReVal Empirical Evaluation Across Benchmarks 0–4

To quantitatively validate cross-sport score parity, identity grounding, adaptive splicing, construct validity, and systems throughput, we benchmarked ReVal across all five MES-5 evaluation splits (scripts/benchmark_reval_engine.py, 5.20 s):

- B0 (Cross-Sport Frequency Normalization, $N_{\text{ref}} = 10.0$): Log-frequency normalization reduces capped cross-sport Coefficient of Variation from 17.97% to 6.75% (−62.44%; Soccer 64.2 → 54.7, Football 94.0 → 64.3, Basketball 99.2 → 58.0) and uncapped impact sum CV from 54.40% to 10.44% (−80.81%).
- B1 (Zero-Shot Multi-Modal Player Attribution Ablation): Across $N = 73$ strict high-stakes Labels-caption.json events evaluated over wide/mixed broadcast feeds (distinct from zoomed crop-level OCR in Track H), fusing Track H OCR (14.63% F1) and Track G ASR/NER (33.33% F1) improves automated F1 to 34.34% (23.29% recall, 65.38% precision; broad $N = 104$: 31.65% F1). On routine Game 0 Half 1 plays ($N = 35$), uncorroborated off-event OCR crops introduce 3 false positives (24.0% Fused vs. 25.53% Track G), highlighting the necessity of high-stakes action gating.
- B2 (Adaptive Acoustic/Optical Splicing): Across 54 cached STFT/MAD profiles (134 high-stakes events), the adaptive splicer improves temporal IoU over fixed $[t-4\text{s}, t+6\text{s}]$ windows from 53.89% to 55.04%, action-apex containment from 85.82% to 100.00%, and clean cut boundary placement from 80.97% to 96.27%.
- B3 (Construct Validity & Sensitivity): Across 10 MES-5 players rated against independent match ratings, ReVal achieves Spearman $\rho = 0.7697$ ($p = 0.00922$) and Kendall $\tau = 0.6000$ ($p = 0.01667$), while preserving 86.0% Top-5 Jaccard stability ($\rho_{\text{MC}} = 0.9957$) over 100 Monte Carlo ±20% weight perturbations.
- B4 (Throughput, A/V Sync & WCAG 2.1): PyAV encodes 250 720p frames (10.0 s clip) in 4.136 s (60.45 FPS, RTF = 0.414), demux/mux audits confirm all 3 rendered reels carry synchronized 44.1/48 kHz AAC audio (has_audio=true, 0.0 s start/duration delta, 21.33–23.22 ms first-packet priming offset $<$ 25 ms), and the #003153/#FF9E1B palette clears WCAG 2.1 AAA on body/badge ink (White 13.43:1, Slate 10.89:1, Navy #001020 9.28:1) and AAA-Large/AA-Normal on $\geq$ 20 px bold Orange accents (6.49:1).

12 Spatial and Geometric Challenges: Completing All 12 SoccerNet Tracks (Tracks I–L)

To encompass the remaining spatial, tracking, and 3D reconstruction challenges on CPU without cloud compute, V5 implements deterministic HSV turf/chalk isolation, Chamfer pinhole DLT calibration, calibrated inverse-homography projection, Kalman+HSV association with cut/dissolve recovery, and Trust-Region Reflective pose optimization in processing/pitch_geometry.py (Table 14). Across Tracks I, J, and L, geometric accuracy is evaluated under reference-anchored pinhole refinement initialized from broad pitch hypotheses (12 keyframes; $N = 92$ GSR players; $N = 82$ depth targets), distinct from unconstrained scratch monocular regression.

Table 14: SoccerNet V5 Spatial and Geometric Tracks (Tracks I–L) on Game 0 Half 1 Broadcast Video.

Challenge Track	Configuration	Primary Metric	V5 Result (vs. Bas
Track I: Camera Calibration	Chamfer Pinhole DLT (12 keyframes)	IoU / AC@10px	0.9601 / 78.47% (vs. 0.09
Track I: Reprojection Error	Chamfer Pinhole DLT (12 keyframes)	Mean / Median Error	8.14 px / 5.34 px (vs. 8
Track J: Game State (GSR)	Calibrated $\mathbf{H}^{-1}$ + 4-Role HSV ($N = 92$)	GS-HOTA / Loc F1@2m	84.78% / 89.62% (vs. 1.91
Track J: Role Classification	Red/Orange/Yellow/Cyan HSV	4-Role Kit Accuracy	98.78% (0.238 m mean p
Track K: Object Tracking	Kalman + HSV + Cut Recovery (650f)	HOTA / IDF1 / IDSW	22.63% / 69.70% / 263 (vs. 68.59
Track K: Player Re-ID	HSV + Scale + Track H ($N = 84$)	Top-1 / mAP	34.52% / 48.08% (vs. 20.24
Track L: 3D Metric Depth	TRF Pose Raycast ($N = 82$, 38.6–94.2 m)	Abs Rel / RMSE / δ_1	0.0496 / 4.455 m / 100.0% (vs. 0
Track L: Hybrid Fusion	Raycast + BBox Regularizer ($N = 82$)	Abs Rel / RMSE / δ_1	0.0512 / 4.513 m / 1

13 Conclusion

We presented the SoccerNet VLM Modernization Benchmark (V2–V5), establishing a rigorous 12-track multi-task foundation spanning semantic video understanding and projective 3D pitch geometry on untrimmed sports broadcasts. Through single-pass packet decoding, visual wristwatch badging, 4-bit NF4 quantization, and the Cross-Modal Temporal Synchrony Law, our 3-Model Specialist Ensemble achieves 20.87% Active Loose-mAP (surging to 24.15% with out-of-sample M_{19} Markov causal rescoring, 0.1840 across all 1,132 MES-5 events under Stage 2 window skipping), 20.00% LOCO cross-validation, and competitive performance across all twelve SoccerNet challenge tracks—spanning Spotting, Captioning, VQA, Foul Recognition, Ball Action Spotting, Replay Grounding, Echoes Entity Resolution, Jersey OCR, Camera Calibration, Game State Reconstruction, Tracking/Re-ID, and 3D Metric Depth—on consumer edge hardware at $0 cloud compute spend.

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